VitalTwin

AI-Powered Clinical Digital Twin

A Living Digital Perspective on Health

Domain: Hypertension Risk (365-day)
Type: Research Prototype
Team: TASK FORCE 141
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Users

- Clinician / Researcher
- Patient (Multilingual View)
- Research Team

1. Client / Presentation Layer (Frontend)

Next.js 16 Web Dashboard (Port 3000)



- Landing Overview
- Digital Twin Workspace
- Cohort Directory
- Early Warning Alerts
- Population Analytics
- ABDM FHIR / CSV Import
- Honest Metrics
- Audit Walk (Cryptographic)
- Dual 3D Visualizer
- Patient Simulator (What-If)
- Multilingual Patient View

Frontend Stack

- Next.js 16
- React
- TypeScript
- Tailwind CSS
- Three.js (3D)
- API Client (lib/api.ts)



HTTPS / REST API (JSON)

2. Backend API Service Layer (FastAPI)

FastAPI (Port 8000)

- **CORS Filter** (Regex Whitelist)
- **Application Router** (app/main.py)
- **Pydantic v2 Schemas** (app/schemas.py)

Core Domain Services

- **Patient Service** (Cohort & 132-Feature Extraction)
- **Model Service** (Inference & Simulation Engine)
- **SHAP Service** (TreeExplainer Attribution)
- **Timeline Service** (Trajectory Drift & Lead Time)
- **Wearable Service** (SSE Telemetry Streaming)
- **FHIR Service** (ABDM FHIR / CSV Import)
- **Audit Service** (SHA-256 Hash Chained Ledger)
- **Visualization Service** (3D + Organ Hotspots)

Backend Stack

- Python
- FastAPI
- Pydantic v2
- Uvicorn
- SQLAlchemy
- Pandas / NumPy

3. Data Sources

- **Clinical EHR Data** (Longitudinal Records)
- **ABDM FHIR / CSV** Import
- **Wearable Telemetry** (Optional)

Validated Patient Data

4. ML Inference Pipeline

Preprocessing (ColumnTransformer) → **Trained XGBoost Model** (XGBoostClassifier) → **Probability Calibration** (Platt Scaling) → **Risk Score Output** (0 - 1) 0.32

5. Explainability & Simulation

- **SHAP Explainability** (TreeExplainer) Top Risk / Protective Drivers
- **What-If Simulation Engine** (Correlated Feature Levers) lever_map.json

6. Storage Layer

- **Relational Database** (Patient Data, Features, Audit Logs)
- **Model Artifacts** (Trained Model, Scaler, Calibrator, lever_map.json)

7. Security & Governance

- HTTPS & CORS Protection
- Request Validation (Pydantic)
- SHA-256 Audit Ledger
- Access Control
- No Direct Patient Identifiers
- Research Use Only

8. Outputs / Visualization

- Risk Score & Confidence Interval
- SHAP Feature Attributions
- What-If Comparison
- **3D Digital Twin** (Heart, Brain, Kidneys, Arteries)

End-to-End Data Flow (Simplified)

1. User selects patient
2. Frontend requests patient data
3. Backend validation & preprocessing
4. Model inference & calibration
5. SHAP explanation (optional)
6. Results visualized in dashboard